Intelligent Ground Vehicle Competition 2026

AutoNav & Self-Drive

Cornell Electric Vehicles

Cornell University

Design Report

[Vehicle Name]

[Vehicle Photo / Sketch]

[Faculty Advisor's Signed Statement of Integrity]

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Contents

1. System Requirements and Design

Things to Write

- Describe the system engineering process you followed to identify important System and Subsystem requirements.
- List at least 2 requirements for: (a) the mechanical design of your vehicle, (b) safety, (c) the electrical/electronic components
- For each challenge you are targeting (AutoNav and SelfDrive), list at least two requirements that guided your design for each of: (a) perception, (b) driving logic, and (c) key performance indicators.
- For each requirement you listed (12 for one course, 18 if both) explain what in these rules elsewhere that drives this requirement, describe how you will quantifiably measure the requirement, and set a target value.

2. Mechanical Design

Things to Write

- Describe the mechanical design of your robot, including CAD or other drawings of your vehicle
- Describe significant decisions on the frame structure, housing, and/or overall design
- Describe significant components of your vehicle's drive system, electrical power system, and the control system that enables computer-controlled motion. Components acquired ready-made must be identified, but their internal components need not be described in detail.
- Describe the significant mechanical components, including the frame, structure, suspension, housing, and weather proofing. Clearly identify the aspects of your mechanical design that are driven by each mechanical system requirement you listed.
- For each mechanical requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

3. Safety

Things to Write

- Describe the robot's safety aspects while being transported, parked, and charging.
- Describe the robot's safety aspects while operating on the course(s), including the mechanical and wireless ESTOP systems.
- Clearly identify the aspects of your safety efforts that are driven by each safety requirement you listed.
- For each safety requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

4. Electrical/Electronic Design

Things to Write

- Describe significant power, electrical, and electronic components in your vehicle, including electronics, computers, sensors, and motor controllers
- Describe significant decisions made in designing/selecting the electrical and electronic components
- Describe the power distribution system's capacity, max. run time, recharge time, and safety
- Clearly identify the aspects of your electrical and electronic components that are driven by each electrical/electronic components system requirement you listed.
- For each electrical/electronic components system requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

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5. Perception

Things to Write

- If you are targeting both challenges, you must include separate discussions of how your perception system matches each course.
- Describe processing of the sensor data to identify course features and items around the vehicle, including obstacles and lane markings, and the use of machine vision to perceive the environment.
- If targeting AutoNav, describe how the system identifies lanes and obstacles.
- If targeting SelfDrive, describe how the vehicle senses pedestrians, stop signs, and road obstacles.
- Describe the internal representation of the course built from the perception data which is used by the driving logic. This might be a sophisticated internal representation of the environment or a simple list of features. Include a discussion of how it is created and updated as the robot moves.
- Clearly identify how your design of the sensor data processing system was driven by the perception requirements you listed.
- For each perception requirement you listed, compare the target value with the most recentlymeasured actual value, and discuss what this tells you about your vehicle
- Developed a custom C++ Iterative Closest Point (ICP) node to process LiDAR point clouds for robust spatial mapping.
- Implemented a 3D-to-2D vertical slicing filter (using ROS 2 pointcloud iterators) to compress 3D LiDAR data, isolating course obstacles at specific Z-heights while filtering out ground and ceiling noise to prevent false obstacle detection.

Obstacle Segmentation

The 3D nature of point clouds allows partial understanding of the geometry and spatial positioning of both static and dynamic obstacles around the vehicle. Specifically, the on-vehicle RoboSense Helios 32 LiDAR provides reliable depth measurements and information of non-occluded obstacle surfaces, constrained to the vehicle's 360 field of view. As a result, these LiDAR scans will be used as the primary source of data for occupancy grid and cost map generation for the vehicle's local and global planning algorithms.

RoboSense Helios 32 LiDAR scans are processed in real-time as raw points in PointCloud2 format and converted into Nav2 costmaps as input for the vehicle's planning algorithms.

Point cloud clustering enables the processing of raw 3D LiDAR data into per-obstacle representation that better captures the spatial properties of obstacles around the vehicle. Clustering is achieved by running either naive Euclidean distance or two layer graph clustering.

Given a raw 3D LiDAR scan, the Euclidean cluster algorithm iterates through all points of the input point cloud and will either start a new cluster to grow or continue to add new neighboring points to the current cluster that is being grown. This algorithm will initialize a cluster or array of 3D points by sampling uniformly at random a point from the set of unclustered seeds. Then, through traversal of a KD-tree containing all 3D points, nearest neighbors of the currently explored point will be added to the cluster if the Euclidean distance between explored and newly discovered points are within a pre-defined threshold. To account for the inverse relation between point density and distance from LiDAR origin, this distance threshold scales with how far the explored point is from the LiDAR origin. Only unclustered points can be added to the currently grown cluster, and only when there exists no more unclustered points that satisfy the Euclidean distance condition will the cluster's growth be complete. New clusters will be initialized until there are no more unclustered point from the original raw 3D LiDAR scan. Given that this is a sequential process that depends on the structure of existing clusters, full parallelization is difficult.

Two layer graph clustering takes advantage of the vertical and horizontal angle resolutions of LiDAR hardware for optimization and requires two separate segmentations of the point cloud. The first segmentation involves clustering points within a threshold along the same horizontal ring – rays projected from LiDAR origin to points belonging to the same horizontal ring possess the same vertical angle from the ground plane. A range graph consists of range graph nodes that contain information about the resulting per-horizontal ring clusters: including the corresponding horizontal ring index and the leftmost and rightmost points of that cluster. The construction of the range graph can be parallelized across horizontal rings. Given that the first segmentation already encodes some understanding of horizontal separations between clusters, the second segmentation involves performing BFS on this range graph and merging nodes that are within a pre-defined vertical distance threshold. Despite having to search through all permutations of the range graph for accurate merging, the number of nodes to search through in the range graph is significantly smaller than the total number of points in the raw LiDAR scan. This is proposed to be linear runtime.

Given the assumption that clustering outputs accurately captures the state of obstacles around the vehicle, obstacle tracking constructs a cost matrix populated by how well past and current clusters via oriented bounding box overlap. Hungarian Algorithm for max bipartite matching is utilized to generate matchings between past and current clusters that minimizes the total cost of overlapping. These matchings are used to update current clusters with their corresponding past clusters' clusterid.

6. Driving Logic

Things to Write

- If you are targeting both challenges, you must include separate discussions of how your driving logic matches each course.
- Design of the lane following and obstacle avoidance systems must be specifically described, along with how the vehicle navigates through its environment. Describe how the system uses GPS for waypoint navigation and localization.
- If targeting AutoNav, describe how your robot uses the perception system's output and other information to follow lanes, detect and avoid obstacles, climb the ramp, and navigate to the GPS waypoints. Include how vehicle motion is generated and monitored to move the vehicle through the course, including

how it deals with complex obstacles including switchbacks and center islands dead ends, traps, and potholes.

- If targeting SelfDrive, describe how the vehicle uses the perception system's output and otherinformation to follow the rules of the road while navigating the course that includes pedestrians,stop signs, road obstacles, and other items.
- Clearly identify how your design was driven by the driving logic requirements you listed.
- For each driving logic requirement you listed, compare the target value with the most recentlymeasured actual value, and discuss what this tells you about your vehicle

7. Key Performance Indicators

Things to Write

- If you are targeting both challenges, you must include separate discussions for each course.
- Describe how you will measure each of the key performance indicators you selected to check that your robot is competitive on the target course(s), what value/result you are targeting, and how you selected that value/result for the KPI.
- For each KPI requirement you listed, compare the target value with the most recently measured actual value, and discuss what this tells you about your vehicle

8. Analysis of Complete Vehicle

Things to Write

- Describe lessons learned during construction and system integration
- Identify key mechanical and electrical components that have failed during testing and discuss what changes you made to prevent repeat failures. Did your changes prevent repeat failures?
- Describe your software testing, bug tracking, and version control processes
- Describe your simulation-based testing process (e.g., Gazebo, etc.)
- Describe physical testing to date (mechanical, electronic, indoors, real world, etc.) and identify key differences in actual performance compared to predictions from simulation.

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9. Cyber Security Analysis

Things to Write

- Modern robotic and human driven vehicles are network-connected, contain large amounts of software from multiple vendors, and are an inviting target for cyber-attacks.
- List 3 cyber vulnerabilities present in your vehicle and discuss what steps should be taken to harden/remove them before it enters series production and is widely deployed.